Ma Toan Bach

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Education

Seneca Polytechnic — Toronto, ON
Bachelor of Engineering in Software Engineering — *Sept 2022 – August 2026*

Experience

Research Assistant — May 2025 – Present
Seneca Applied Research — *Toronto, ON*

- Designed and deployed an automated firmware unit-test generation workflow using Python, LangChain/LangGraph and FastMCP, cutting manual setup time by 90%.
- Architected and containerized a microservices test-generation FastMCP server to expose 12+ test tools via an internal API platform, enabling parallel and sandboxed test runs with robust concurrency controls.
- Implemented a RAG pipeline and a parser that indexed over 4,000 code symbols and knowledge-base records, boosting unit-test coverage from 84% to 99% while reducing token usage by about 32%.
- Integrated end-to-end observability with OpenTelemetry, Grafana, Tempo and Loki, decreasing debugging time by 30% and improving reproducibility.
- Partnered with AMD firmware engineering teams to refine prompts, guardrails and platform features, improving AI-generated test quality and driving adoption across the organization.

AI Engineer Intern — December 2025 – April 2026
Pathway Communications — *Toronto, ON*

- Built an AI browser automation system to extract and capture information from third-party portals using Playwright, Typescript, OpenCode SDK with sandboxed execution, Jenkins CI/CD and Docker for containerization, reducing manual coding time by 96%.
- Assisted in testing AI automation workflows for ticket processing in production OTRS system using the Google Agent Development Toolkit and Python.
- Built a Django-based observability dashboard to support debugging and testing of AI automation workflows; collaborated with ITSM, DevOps teams and stakeholders to gather feedback, ensuring alignment with project requirements and goals.
- Architected and maintained a FastMCP server platform that exposed database tools across MySQL, PostgreSQL and SQLite for AI agents to query client information.
- Integrated and configured Langfuse, Grafana, Prometheus and OpenTelemetry to monitor LLM agent workflows and MCP server metrics to track cost, latency, token usage and error rates, improving reliability and speeding up troubleshooting.

Certifications

Certified Kubernetes Application Developer (CKAD)	Issued: 2025-01 — Verify
HashiCorp Certified: Terraform Associate (003)	Issued: 2025-10 — Verify
Red Hat Certified System Administrator (RHCSA)	Issued: 2025-09 — Verify
Red Hat Certified Engineer (RHCE)	Issued: 2026-04 — Verify

Technical Skills

Languages: Python, Go, TypeScript, SQL
DevOps: Linux, RHEL, Proxmox, Bash, Docker, Kubernetes, Ansible, Terraform, GitHub Actions, Jenkins
Observability: Langsmith, Langfuse, Grafana, Prometheus
Cloud: AWS (EC2, EKS, ECR, IAM, S3, RDS)
Full-stack: FastAPI, FastMCP, OpenAPI, REST API, gRPC, Gin, React, Tailwind CSS, HTML
Databases/AI: LangChain/LangGraph, Google ADK, ChromaDB, PostgreSQL, MongoDB, DynamoDB, OpenAI models

Projects

datatrace | *TypeScript, React, Next.js, AWS (EKS/ECR), Docker, Kubernetes, GitHub Actions* *Link, GitHub*

- Collaborated in a 5-person hackathon team to prototype a food-industry analytics + POS dashboard for inventory, sales trends, and near-expiry alert workflows.
- Led the MVP front end in Next.js/React, shipping company vs. supplier views, KPI dashboards, certification workflow UI, and PDF report generation.
- Implemented CI/CD with GitHub Actions to build and push Docker images to AWS ECR and deploy the container to Amazon EKS with Kubernetes manifests and ingress.

leetyap.com | *TypeScript, React, Next.js, Redux Toolkit, Tailwind CSS, Auth0, CodeMirror, OpenAI Realtime* *Link, GitHub*

- Built a solo LeetCode-style interview practice platform with an in-browser problem workspace (problem view, Python editor, and execution output) plus timers and settings.
- Integrated an AI voice interviewer using OpenAI Realtime to guide users through a structured interview flow (approach, implementation, testing, and complexity review).
- Implemented authentication with Auth0 and connected client-side code execution via Judge0, deploying the web app to Vercel with GitHub-based deployments.

Homelab Cluster GitHub

- Maintained a 3-node RHEL cluster on a **Proxmox** server as a hands-on lab to explore core Linux system administration and automation.
- Practiced ad-hoc administration—local storage to manage storage, filesystems, LVM, systemd, firewalld, SELinux, SSH, user/group permissions, IPv4/IPv6 networking.
- Explored to set up a lab **Kubernetes** cluster on a **Proxmox** server using **kubespray**.

Seneca Networking Lab | *Cisco IOS CLI, IPv4/IPv6, Cisco Catalyst 2960 and 4221, Packet Tracer* *GitHub*

- Set up and tested small computer networks by connecting switches and PCs, assigning IP addresses, and verifying communication with ping and show commands.
- Configured routers and switches with basic security (passwords, banners) and enabled both IPv4 and IPv6 connectivity to ensure devices communicated end-to-end.

Compiler in Go | *Go, GitHub Actions CI/CD* GitHub

- Created a custom bytecode compiler and stack-based virtual machine (VM) in Go, following Thorsten Ball's *Writing A Compiler In Go*.
- Transformed an existing interpreter into a VM capable of executing compiled code and achieved 3x speed-up over the existing interpreter.
- Designed and implemented compiler and VM components such as custom bytecode instruction definitions, an encoder/decoder with a basic mini-disassembler, and features like symbol tables, a constant pool, function and jump handling, built-ins, and closures.

Very Simple Bank | *Go, Gin, gRPC, AWS, Docker, Kubernetes, GitHub Actions, PostgreSQL* *GitHub*

- Implemented back-end APIs for a banking system, enabling users to create accounts, authenticate securely and perform money transactions using **Go**, **Gin** and **gRPC**.
- Wrote unit tests to cover 57.1% of the codebase, improving reliability and ensured seamless **CI/CD** with **GitHub Actions**.
- Built, tagged and pushed the back-end images to **Amazon ECR** using **Docker**, and deployed them to **Amazon EKS** with **Kubernetes**.

Job Scraper | *JavaScript, Node.js, Playwright, OpenAI APIs, DynamoDB, Playwright* *GitHub*

- Automated a daily internship job scraper that scans LinkedIn for the latest postings.
- Leveraged **Node.js + Playwright** to navigate dynamic webpages in Chromium for accurate data extraction.
- Used **ChatGPT APIs** to categorize and filter unaligned postings, ensuring relevant job results.
- Checked **DynamoDB** to avoid duplicates; assigned a two-week TTL for entries and broadcasted new jobs to two Discord channels, supporting **144 students**.

HTTP Server | *C++, TCP, HTTP, multithreading, Shell, CMake* *GitHub*

- Designed and implemented a **HTTP** server using **C++** and **CMake**, handling GET/POST requests, serving static files and **gzip** compressed files, and managing multiple concurrent connections
- Extracted URL paths and implemented routing logic to serve different resources and applied **multithreading** to optimize server performance to handle concurrent connections efficiently

DNS Server | *C++, UDP, DNS protocol, Shell, CMake* *GitHub*

- Built a **C++** DNS server to parse and respond to basic queries over **UDP**, simulating real-world DNS on a Unix-based system.
- Created a custom packet-parsing algorithm with **domain name compression** (per **RFC 1035**) to reduce packet size and boost efficiency.
- Used **dig CLI tool** for testing and verified functionality by returning a default IP address (`8.8.8.8`).